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Properties of Materials
Lecture 6

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Lecture 4: Availability of Materials, Case Study, Exam Question 2013, Example Questions, The World of Materials (2), Materials Selection in Design (2), Properties of Engineering Materials (3), Three-Way Relationship (2), Learning Objectives, Why Do We Need a Selection Methodology, Mechanical Properties (3), Thermal Properties (3), Trade-Offs Must Be Made, Materials Selection in Design (3), Materials Selection Process (1)
Lecture 5: Step 1 - Translation, Step 2 - Screening, Materials Selection Process (2), Step 3 - Ranking Materials, Example - Hydraulic Cylinder Tie Rods, Step 4 - Refining Selection, Concepts and Terminology (2)
Lecture 6: The Elastic Moduli, Deformation of Brittle Materials, Deformation of Ductile Materials, Hooke's Law - Young's Modulus, Linear Elastic Behaviour, Structural Engineering, What Might Have Saved the Bridge, Engineering Stress and Strain, True Stress and Strain, Comparison

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The elastic moduli

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Recap from last time…

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Why do we need a selection methodology?

- Material properties vary enormously:
  - e.g. Order of magnitude difference between weakest and strongest ferrous alloys
- Multiple factors complicate design choices:
  - **Function** – loading, environment, design life etc.
  - **Cost** – availability of raw materials and manufacturing processes
  - **Time** – production deadlines and required manufacturing volumes

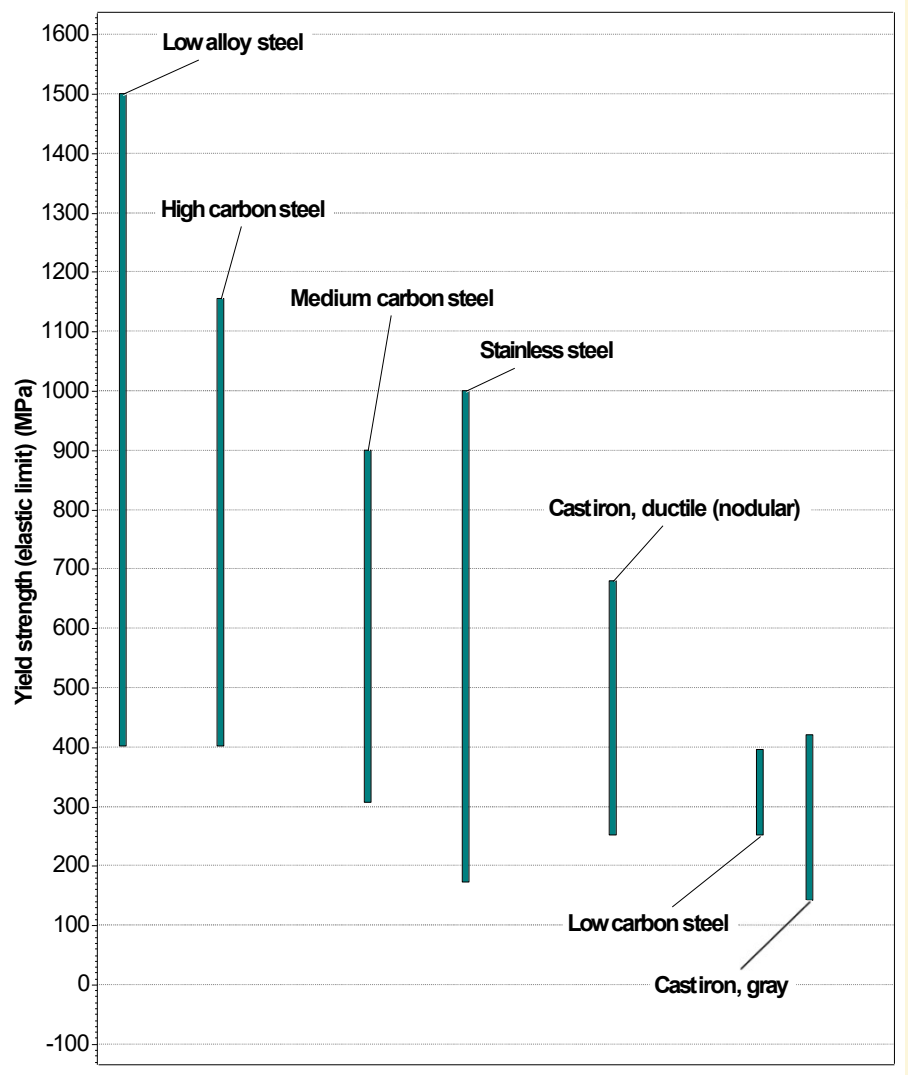


Image created using CES Edupack 2011

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Griff on the scales

https://www.flickr.com/photos/castaway_in_wales/6182582594/

Trade-offs must be made!

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Materials selection process (1)

1 – Translation	• Express design requirements as **constraints** and **objectives**
2 – Screening	• **Eliminate materials** based on constraints and objectives
3 – Ranking	• **Rank materials against metrics** describing best performance
4 – Supporting information	• Use additional information to **refine selection**

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Step 1 – Translation

Design requirements

Function	• What does the component do?
Constraints	• What essential conditions must be met?
Objectives	• What needs to be maximised or minimised?
Free variables	• Which design variables are free?

Data from: Ashby, M. F., (2011), *Materials Selection in Mechanical Design*

Material	Cost (£ kg^{-1})	Density (kg m^{-3})	Failure strength (MPa)	Index, *M*
Medium carbon steels	0.44 – 0.48	7800 – 7900	305 – 900 (yield)	**Best: 4** **Worst: 12**
Aluminium alloys	0.98 – 1.11	2500 – 2900	30 – 500 (yield)	**Best: 5** **Worst: 107**
Carbon-fibre reinforced plastic	26 – 29	1500 – 1600	550 – 1050 (tensile)	**Best: 37** **Worst: 84**
Bamboo	0.91 – 1.37	600 – 800	35 – 44 (tensile)	**Best: 12** **Worst: 31**

Example – Hydraulic cylinder tie rods (5)

- Material performance index (**smaller is better**):

$$M = \left(\frac{C_m \ \rho}{\sigma_f}\right)$$

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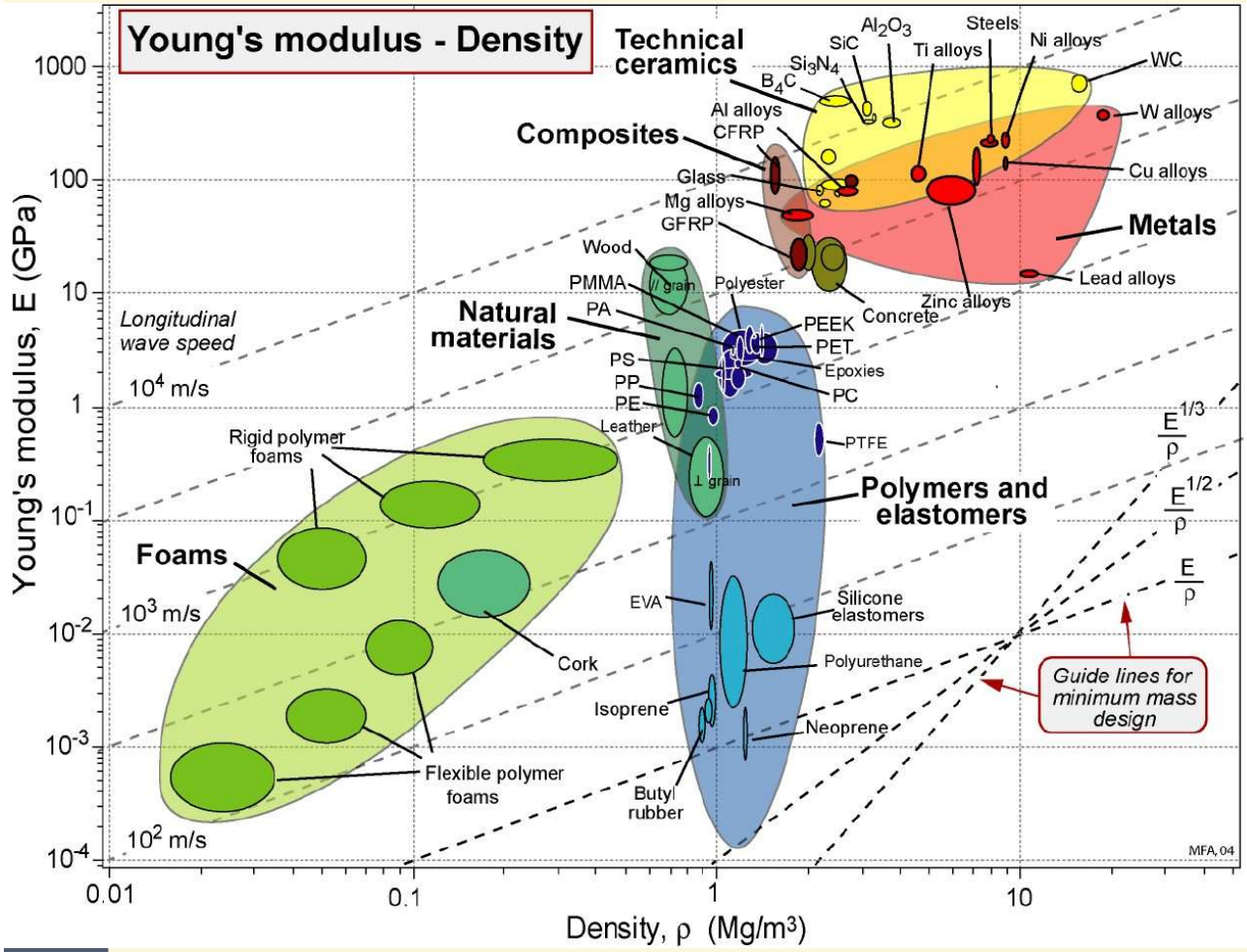
Learning objectives

- Review **stress-strain curves** for ideal brittle and ductile materials

- Understand the origins of **linear elastic behaviour**, from Hooke's Law to Young's Modulus and the 'ball and spring' model

- Learn and appreciate the difference between **'Engineering' stress/strain** and **'True' stress/strain**

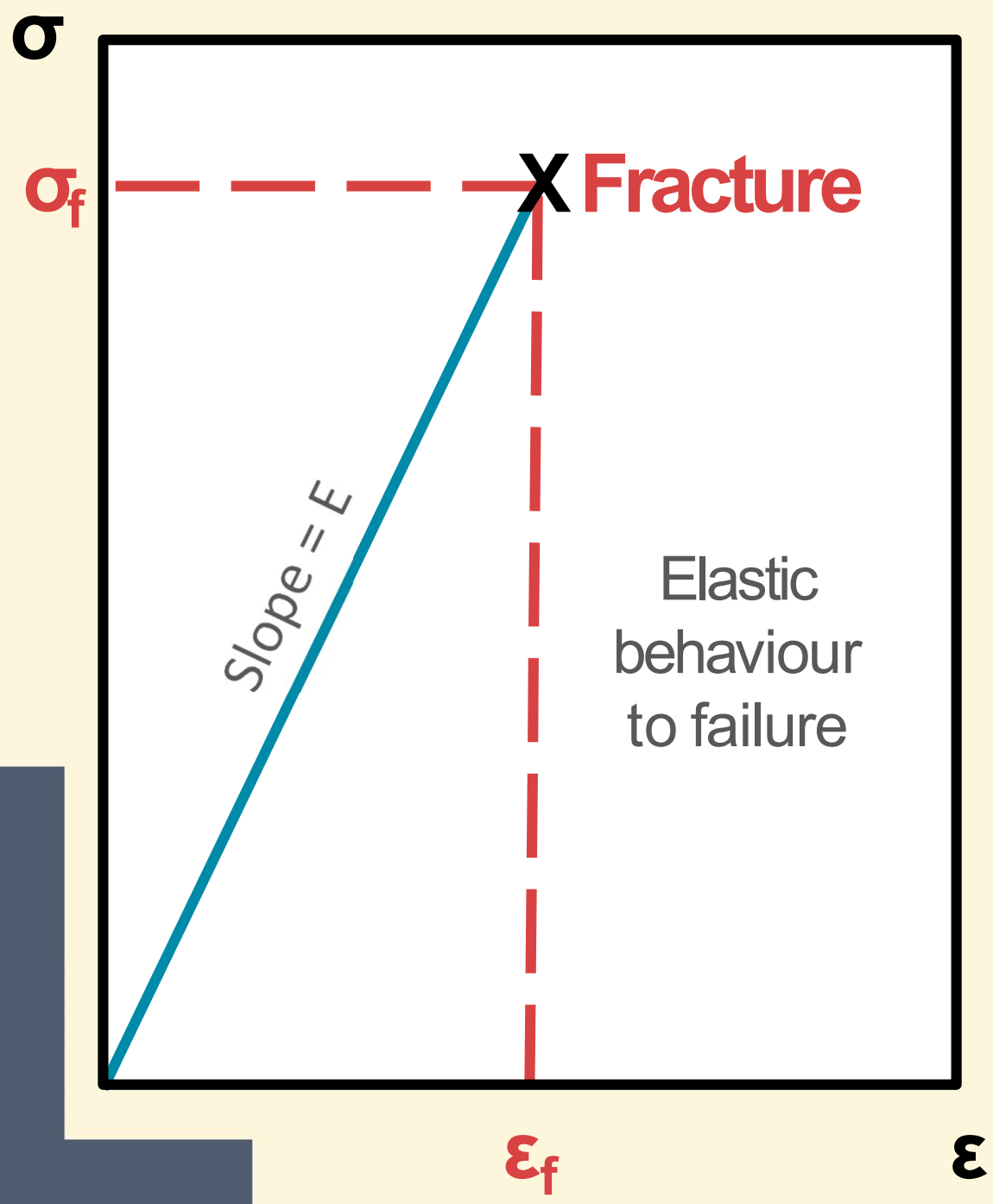
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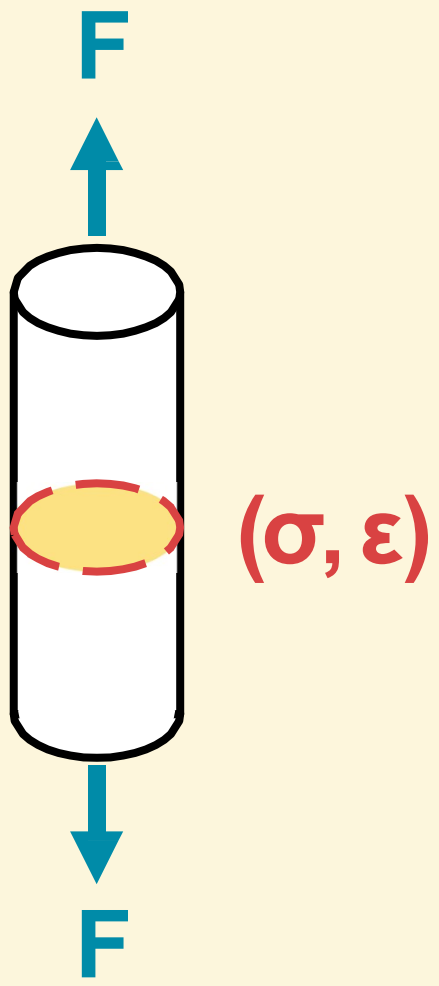
Mechanical properties

- Influencing factors:
  - **Atoms**
  - **Bonds**
  - **Crystals**
  - **Grain structure**
- Manipulated by:
  - Heat treatments (alloys)
  - Making composites
  - Making foams

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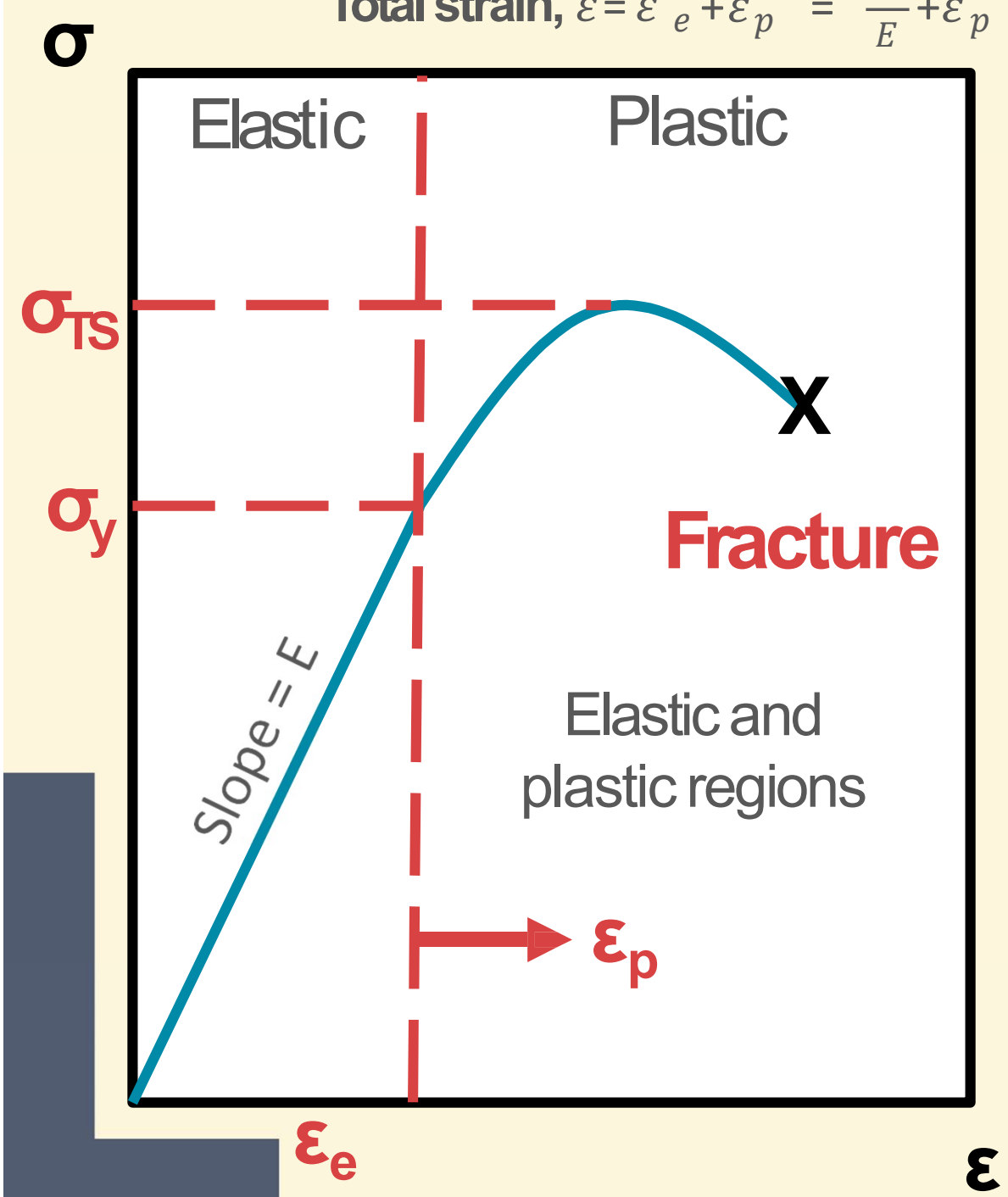


Deformation of brittle materials

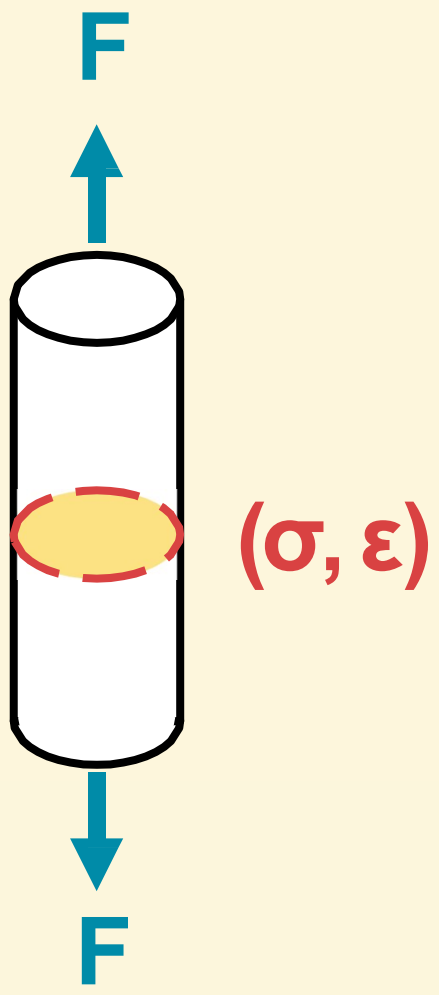


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Total strain, $\varepsilon = \varepsilon_e + \varepsilon_p = \frac{\sigma_y}{E} + \varepsilon_p$



Deformation of ductile materials



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Hooke's Law, Young's Modulus…

…*and the longest derivation in history!*

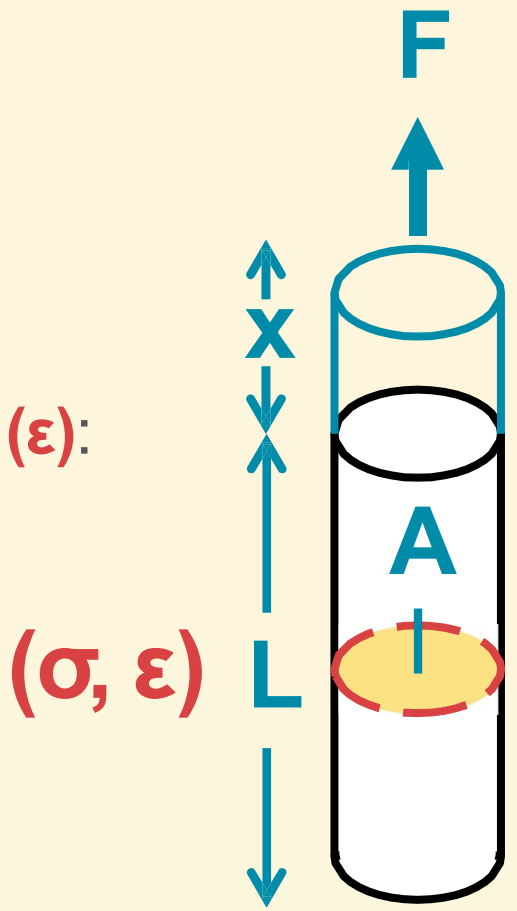
- Hooke's Law (1678): *"ut tensio, sic vis" - Hooke, 1660*
  - Force (F) = Stiffness (k) ×Extension (x)

- Divide through by Area (A) to develop **Stress (σ)**:
  - $\sigma = \frac{kx}{A}$

- Multiply and divide by Length (L) to develop **Strain (ε)**:

$$-\sigma = \frac{kL}{A}\varepsilon \Rightarrow \frac{kL}{A} = \frac{\sigma}{\varepsilon}$$

- Young's Modulus (1807):
  - Elastic Modulus (E) = $\frac{\text{Stress}(\sigma)}{\text{Strain}(\varepsilon)}$



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Linear elastic behaviour

- Ball and spring model of elasticity:
- Material in tension –bonds stretched and atoms move apart
- Material in compression –bonds compressed and atoms move closer together

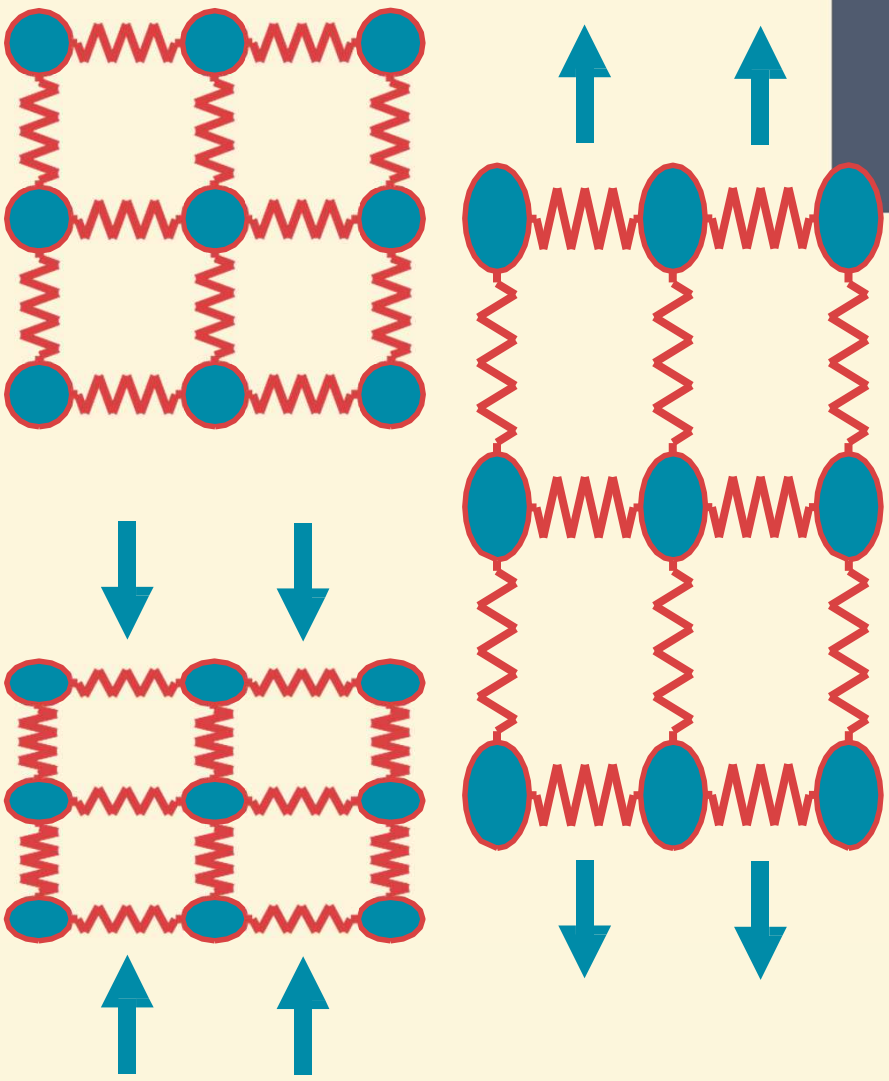


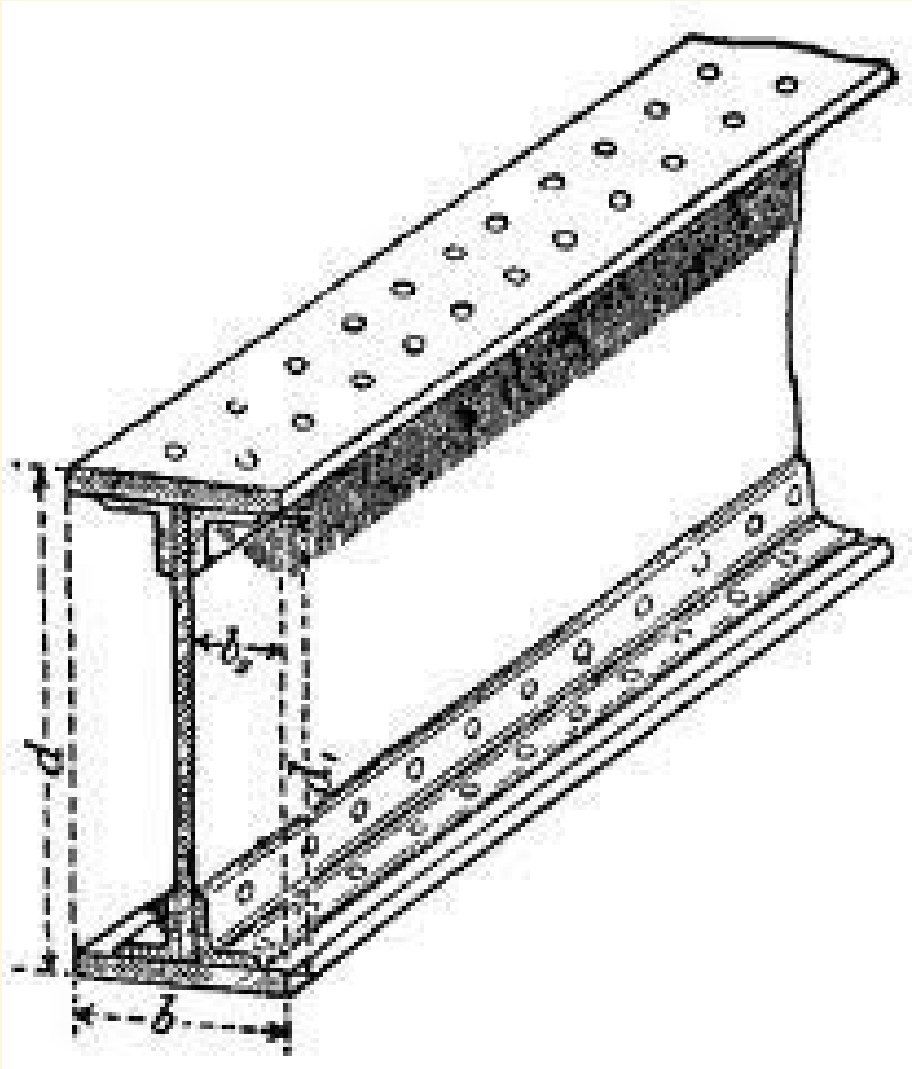
Illustration adapted from: Gordon, J.E., (1978), *Structures or Why Thing's Don't Fall Down*

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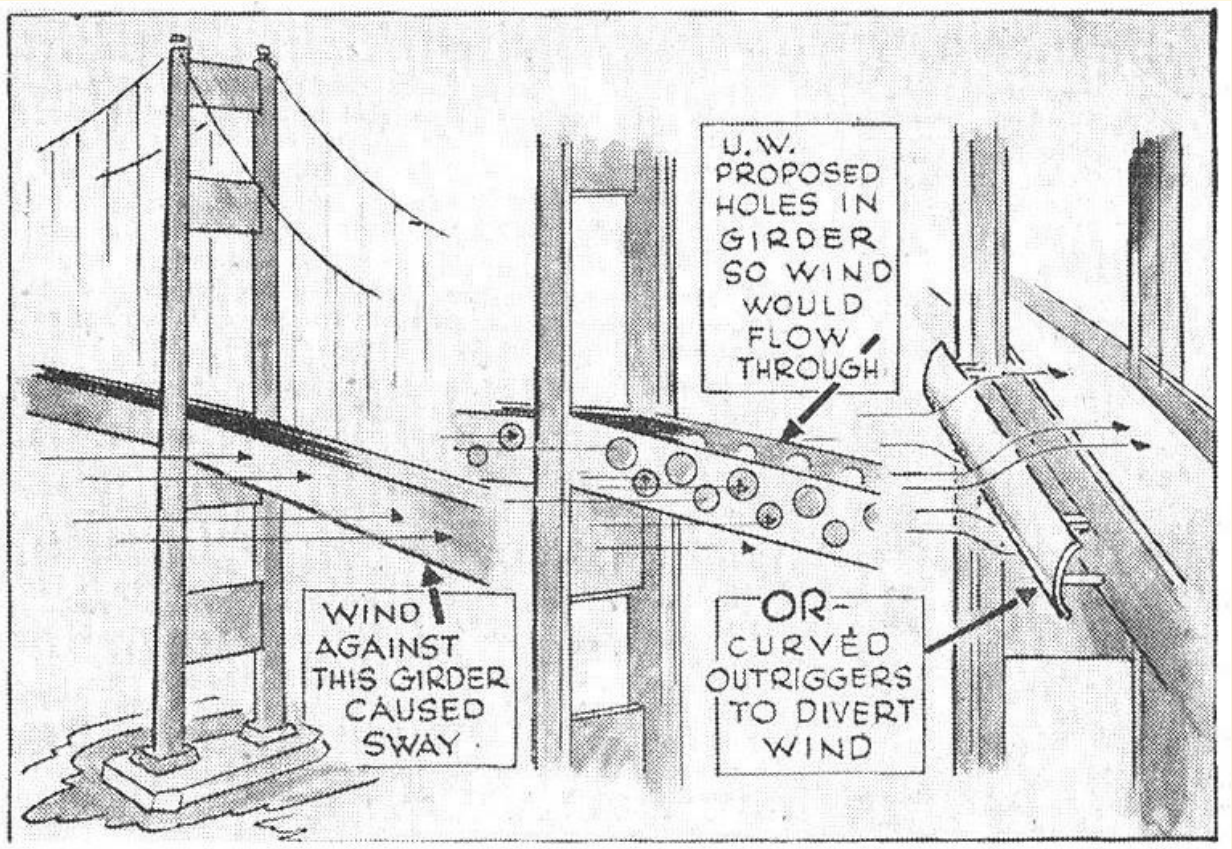
Structural engineering (c.1940)

- Tacoma Narrows Bridge, Washington:
  - 5939ft (1810m) long
    2800ft (853m) span
  - 39ft (12m) wide

- Narrow girders:
  - Solid I-beam section steel
  - 8ft (2.4m) deep



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University of Washington engineers made a test Saturday on their $14,000 model of The Narrows Bridge, attempting to eliminate the dangerous wind sways which finally caused the real-life structure to collapse yesterday. The sketch at left shows the flat horizontal girder which offered resistance to winds, causing the sway. University recommendations were (center) to drill holes with a torch in the girder, permitting the wind to pass through; or (right) to erect an $80,000 streamlined buffer alongside the girder, to divert winds. Their tests showed the latter materially reduced the vibrations, might have saved the bridge.

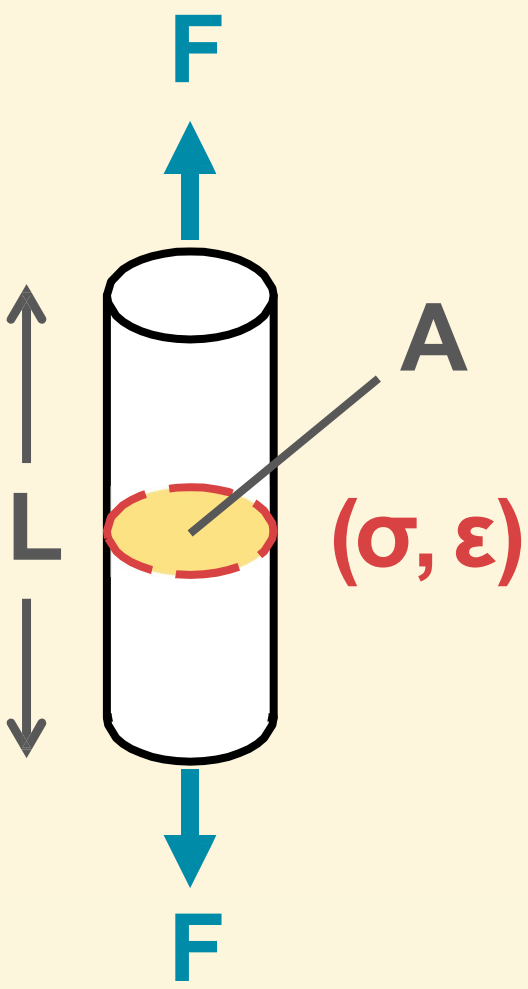
What might have saved the bridge?

- **Insufficient stiffness** in the direction of the cross-winds…
- …alternatively the design was such that the **drag** on the cross-section was too high

Engineering stress and strain

- In uniaxial tensile testing, the applied force creates a stress that results in strain (elongation) of the material
  - Stress, $\sigma_n = \frac{\text{Force}}{\textbf{ORIGINAL} \text{ area}} = \frac{F}{A_0}$
  - Strain, $\varepsilon_n = \frac{\text{Change in length}}{\textbf{ORIGINAL} \text{ length}} = \frac{\Delta L}{L_0}$

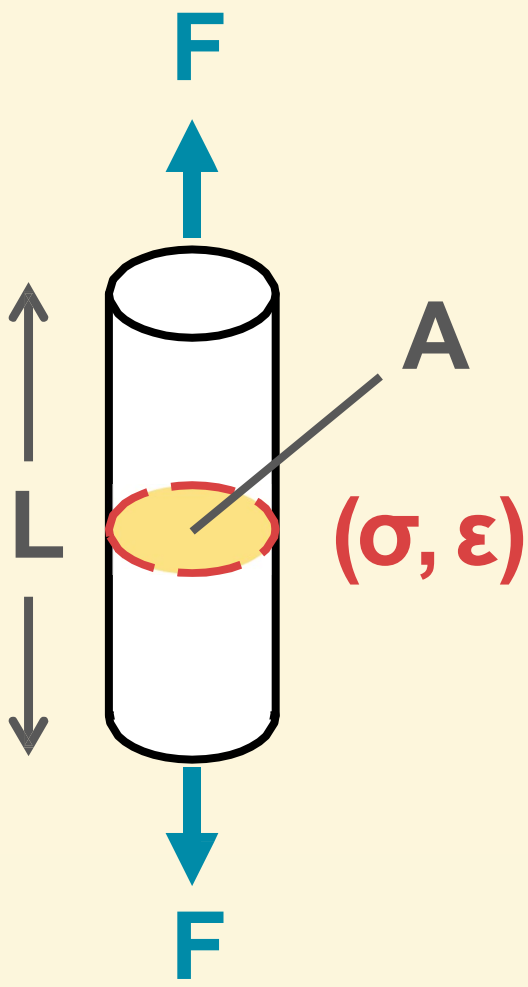
- Engineering or **nominal** stress/strain calculated from the **original dimensions** of the part



True stress and strain (1)

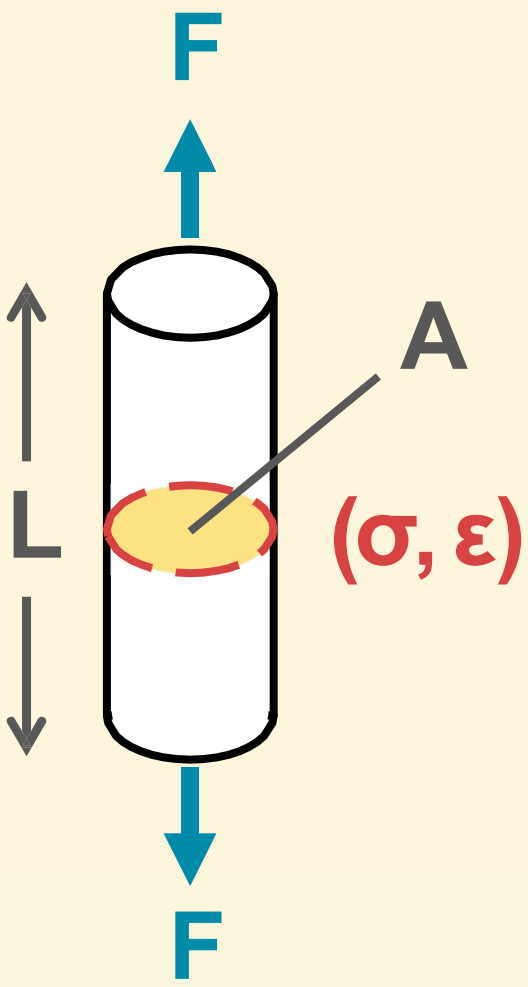
- True tensile strain is given as:
  - Strain, $\varepsilon_t = \int_{L_0}^{L_i} \frac{dl}{l} = ln\frac{L_i}{L_0}$

- As:
  - $\varepsilon_n = \frac{L_i - L_0}{L_0} \Rightarrow \frac{L_i}{L_0} = 1+\varepsilon_n$
  - $\boxed{\varepsilon_t = \ln(1+\varepsilon_n)}$



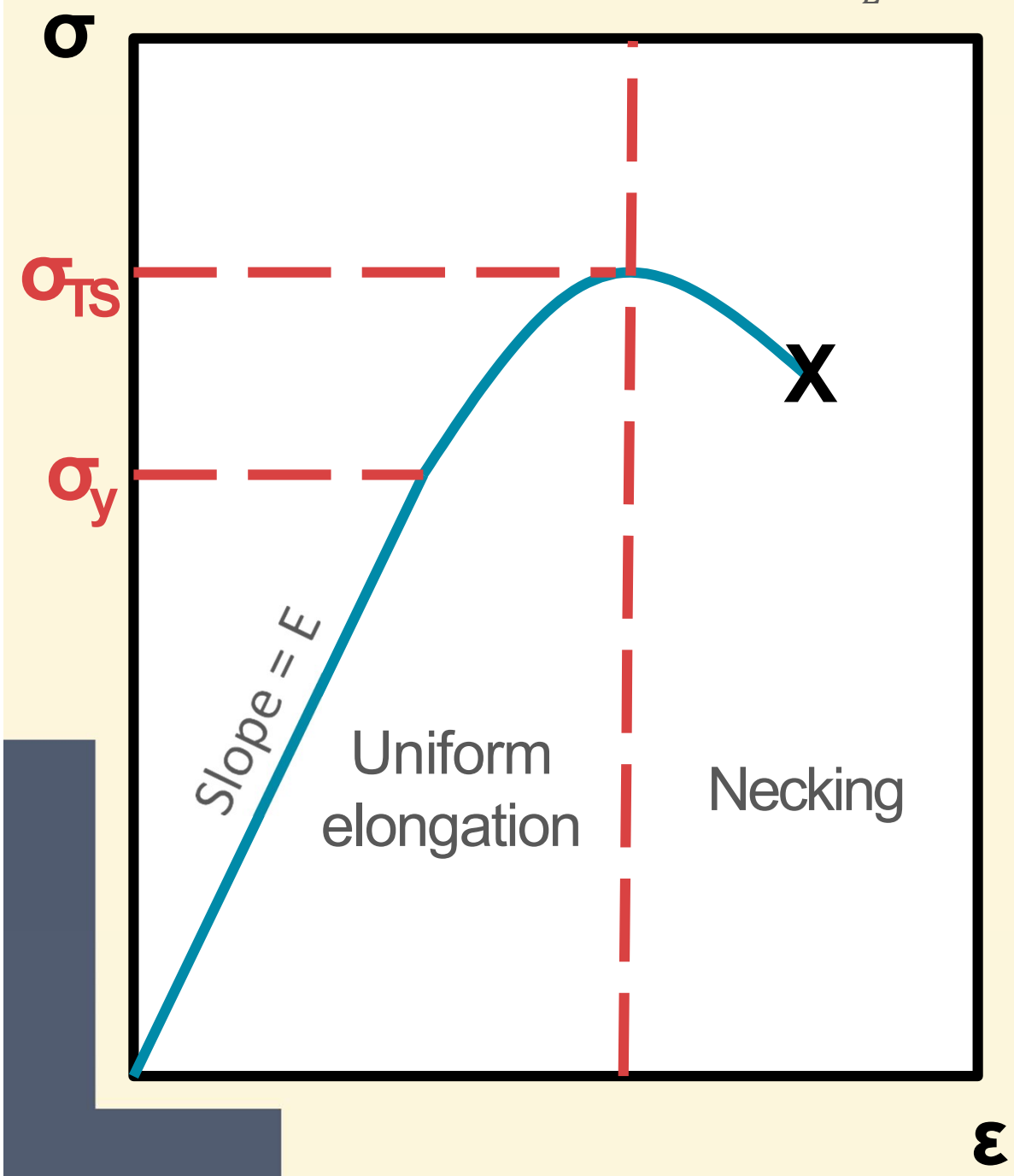
True stress and strain (2)

- True stress is calculated from the **instantaneous area** at any load:
  - Stress, $\sigma_t = \dfrac{\text{Force}}{\text{INSTANTANEOUS area}} = \dfrac{F}{A_i}$
- Neglecting volume change during plastic deformation:
  - $\dfrac{A_i}{A_0} = \dfrac{L_0}{L_i}$
  - $\sigma_t = \dfrac{F}{A_i} = \dfrac{F L_i}{L_0 A_0} \Rightarrow \sigma_t = \sigma_n \left(\dfrac{L_i}{L_0}\right)$
  - $\boxed{\sigma_t = \sigma_n(1 + \varepsilon_n)}$

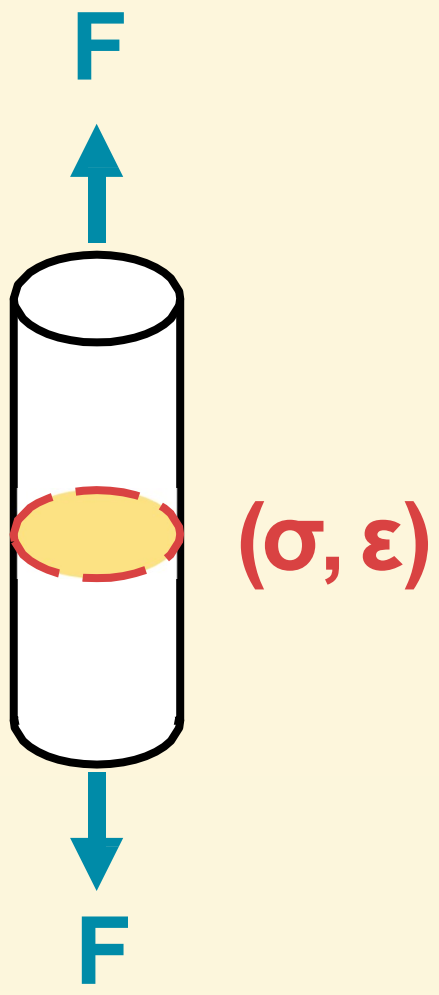


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Total strain, $\varepsilon = \varepsilon_e + \varepsilon_p = \frac{\sigma_y}{E} + \varepsilon_p$



Deformation of ductile materials



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Comparison

- Engineering curve:
  - Drop in stress due to rapid onset of necking
  - Larger scale necking follows
- True curve:
  - Continuous increase in stress and strain
  - Instantaneous values considered

$$\varepsilon_t = \ln(1 + \varepsilon_n) \qquad \sigma_t = \sigma_n(1 + \varepsilon_n)$$

